判斷 logic

- mean
  - single → t 雙
  - two → paired-t 雙
  - multi → ANOVA 單(右)
- proportion
  - single → z 雙
  - two → z 雙
  - multi → chi-square 單

Jury Decision	Reality: H_0 true: defendant did not commit crime	Reality: H_0 false: defendant did commit crime
Reject H_0: find defendant guilty	Type I Error ("false positive")	Correct decision
Do not reject H_0: find defendant not guilty	Correct decisions	Type II error ("false negative")

	Sample Statistic	Population Parameter
Mean	$\bar{x}$	μ
Standard deviation	s	σ
Proportion	p	π

Std error

Sample mean formula: $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$

$$Var(\bar{X}) = Var(\frac{1}{n}\sum_{i=1}^{n} X_i) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{n}{n^2}\sigma_X{}^2 = \frac{\sigma_X{}^2}{n}$$

Reminder: $Var(aX) =$

$$SE(\bar{X}) = \sigma_{\bar{X}} = {}^{\sigma}/_{\sqrt{n}}$$

Std^2 = Var

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

CI: point estimate ± margin of error(E)

- mean → $CI = \bar{X} \pm z^*(\frac{\sigma}{\sqrt{n}})$

  $\downarrow$ $z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$

  $(n-30)$

- σ未知時 → $CI = \bar{X} \pm t_{n-1}(1-\alpha)(\frac{s}{\sqrt{n}})$

- proportion → $CI = p \pm z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$

ONE SAMPLE

$H_A = \mu \gtrless \mu_0$

T-Test $T = \frac{\bar{X}-\mu}{s/\sqrt{n}} = \frac{\text{observed-hypo}}{\text{std error}}$

z-test $z = \frac{p-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{\text{sample-hypo}}{\text{std error}}$

Two SAMPLE

$H_A = \mu_1 \lessgtr \mu_2$

T-test $T = \frac{X_1 - X_2}{\sqrt{\frac{s_1^2}{n_1} \cdot \frac{s_2^2}{n_2}}}$ ddof: n_1-1, n_2-1

Paired T-test $T = \frac{D}{s/\sqrt{n}}$

z-test $z = \frac{p_1 - p_2}{\sqrt{p_{all}(1-p_{all})(\frac{1}{n_1}+\frac{1}{n_2})}}$

MULTI SAMPLE

$H_A = \mu_1, \mu_2, \mu_3$ 其中一个

ANOVA

$$F = \frac{MSG}{MSE} = \frac{\text{mean between group}}{\text{mean square error}}$$

$$MSG = \frac{\sum_{i=1}^{k} n_i(\bar{X}_i - \bar{X})^2}{k-1}$$ ← ddof (k group)

$$MSE = \frac{\sum_{i=1}^{k}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_i)^2}{n_i - k}$$ ≒ sample的次種平均

↖ ddof

Chi-square

$$\chi^2 = \frac{(O-E)^2}{E} = \frac{(\text{observed} - \text{expected})^2}{E}$$

Chi-Square Test Type	Purpose	Example
Goodness-of-fit test	Compares observed vs. expected frequencies for one categorical variable	Checking if customer preference for product colors matches expected proportions
Test for independence	Checks if two categorical variables are associated	Does education level influence voting behavior?
Test for homogeneity – 敦好	Compares distributions of a categorical variable across multiple groups	Do regions have the same preference for mobile phone brands?

- profolio allocation = $\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$ where $\sigma_X^2 = Var(X)$, $\sigma_Y^2 = Var(Y)$, and $\sigma_{XY} = COV(X,Y)$

LEXICAL (write)

SELECT ✓
FROM ✓
WHERE ✓
GROUP BY
HAVING ✓
UNION ✓
ORDER BY ✓

LOGICAL (execute)

FROM
WHERE
GROUP BY
HAVING
} create table
SELECT — extract
DISTINCT
UNION
} duplicate merge
ORDER BY — order

- Subquery ✓ X GROUP BY / WINDOW FUNCTION

① return single row / single column.
→ use in equality / inequality / comparison.
= != >, >=

② return single row / multi column
→ use in IN / NOT IN

③ return multi columns.
→ 列出 steps 一個個拼湊.

- WHERE

Equality / Inequality:
Range = WHERE ___ BETWEEN ___
Membership = WHERE ___ IN ___ or subqueries.
Matching = WHERE ___ LIKE ___
{ partial string matches — / %.
NULL = WHERE ___ IS NULL

- CASE

CASE
WHEN ___
THEN ___
ELSE ___
END AS ___

- JOIN

SELECT
FROM JOIN ___ ON ___
JOIN ___ ON ___

SELECT
FROM
JOIN
(SELECT ...
)

- UNION

UNION
UNION ALL (有 duplicate row)
INTERSECT
EXCEPT

- Date and time

CURRENT DATE
CURRENT_DATE()
CURRENT_TIME()
CURRENT_TIMESTAMP()
YEAR(), month(), ...

DATE FORMATTING:
TO_DATE(string, format)
DATE_TRUNC() 切到想要的粒度.

JOIN = combine columns. (加左右)
UNION = combine rows (加上下)

- CTE

WITH ___ AS
(
subquery.
)
Main query.

- WINDOW FUNCTION

SELECT ___
WINDOW FUNCTION OVER(
PARTITION BY ___
ORDER BY ___ ___
FRAME ↓
) AS name {RANGE | ROWS} BETWEEN ___ AND ___
FROM

{ n / UNBOUNDED + PRECEDING / FOLLOWING
current row

WINDOW FUNCTION

- RANKING
ROW_NUMBER() 1,2,3,4,5,6,....
RANK() 1,2,3,4,4,4,7,8,9
DENSE_RANK() 1,2,3,4,4,4,5.
PERCENT_RANK()

- extra: NTILE()

- VALUE
LAG = 往前看
LEAD(column, offset, default) 往後看
FIRST_VALUE()
NTH_VALUE()

- AGGREGATE

primary key:
foreign key: unique identity in difference table.
composite key:

ROW
RANGE duplicate 算一行.

Aggregate Functions
- COUNT(): Returns the number of rows that match a specified criterion.
- SUM(): Calculates the total sum of a numeric column.
- AVG(): Computes the average value of a numeric column.
- MIN(): Finds the minimum value in a set.
- MAX(): Finds the maximum value in a set.

String Functions
- CONCAT(): Concatenates two or more strings.
- TRIM(): Removes leading and trailing spaces from a string
- REPLACE(): Replaces occurrences of a specified substring within a string.
- CHARINDEX() / POSITION(): Returns the position of a substring within a string.

Scalar Functions
- UCASE() / UPPER(): Converts a string to uppercase.
- LCASE() / LOWER(): Converts a string to lowercase.
- ROUND(): Rounds a numeric field to the specified number of decimal places
- .LEN() / LENGTH(): Returns the length of a string.
- SUBSTRING() / MID(): Extracts characters from a text field.

Date Functions
- GETDATE() / CURRENT_DATE: Returns the current date.
- DATEPART(): Returns a single part of a date/time, such as year, month, day, etc.
- DATEDIFF(): Returns the difference between two dates.
- DATEADD(): Adds or subtracts a specified time interval from a date.

Numeric Functions
- ABS(): Returns the absolute value of a number.
- POWER(): Returns the value of a number raised to the power of another number.
- SQRT(): Returns the square root of a number.
- MOD(): Returns the remainder of a division operation.